

Importing Slide Files / SSR Analysis

Slide is a 2D limit equilibrium slope stability program produced by Rocscience. *Phase2* version 8 can import files written by version 6 of *Slide*. This allows you to perform a finite element stress analysis and slope stability analysis on a *Slide* model, using *Phase2*.

This tutorial will provide an overview of Slide file import and the Shear Strength Reduction method in *Phase2*, and then demonstrate the procedure with an example.

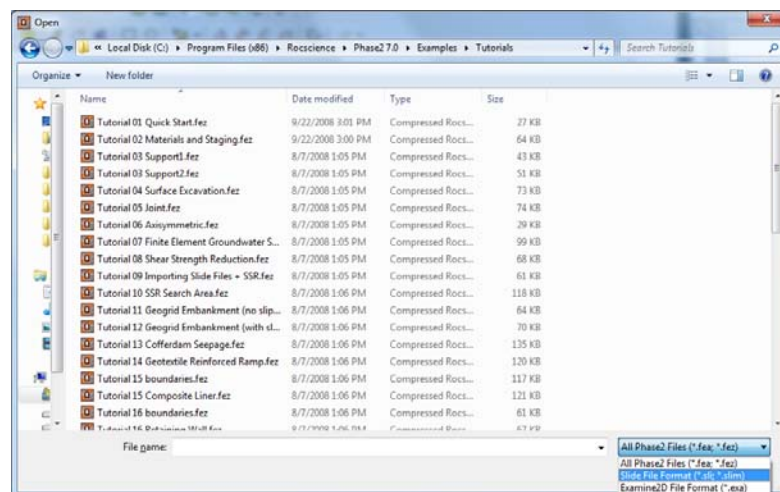
Topics Covered

- Importing a *Slide* file
- *Slide* options which are supported in *Phase2*
- *Slide* options which are not supported in *Phase2*
- Shear Strength Reduction (SSR) analysis

Importing a Slide Data File

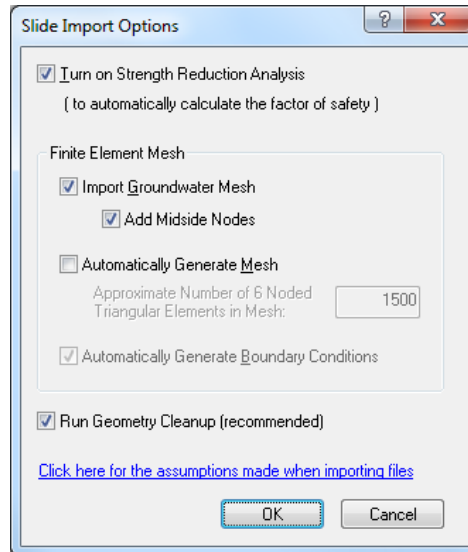
To import a *Slide* data file (.slim file), there are two possible methods:

1. You can use the File > Import > Import Slide File option.
2. Or you can use the File > Open option and set the Files of Type to Slide File Format (*.slim) as shown below.



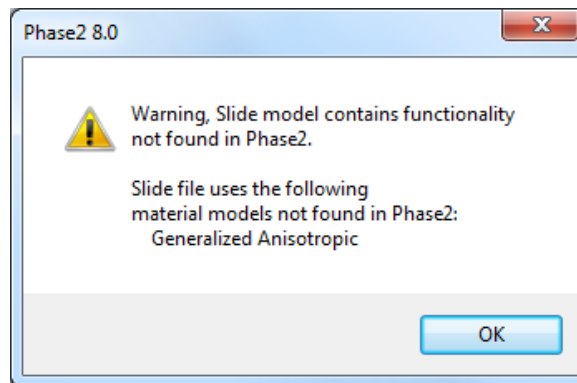
Both methods provide identical functionality for importing *Slide* data files into *Phase2*.

After selecting the *Slide* data file that you wish to import, you will see a dialog with options pertaining to how you wish to import the file.



In general, you will simply press OK, but there might be instances where you wish to modify boundaries, customize the mesh, or not start by running a Shear Strength Reduction (SSR) analysis to determine the factor of safety of your slope. In which case, you can use this dialog to customize how the *Slide* file is imported.

After the import, you might see a warning dialog such as:



Not all functionality in *Slide* is supported by *Phase2*. Certain material and support models are not supported (see below). If a *Slide* model contains unsupported functionality, a warning dialog is issued. In this case, the user must change the material or support models to one supported by *Phase2*. The method for defining material and support models is very similar between *Slide* and *Phase2*, so the user should have no problem changing the model.

Shear Strength Reduction Method

The Shear Strength Reduction (SSR) method is widely used to determine the factor of safety of a slope using the finite-element method and is used in *Phase2* version 8. The method simply reduces the shear strength of the material until the model becomes unstable. The point of instability is taken as the factor of safety of the slope. It is not the purpose of this document to describe the method. However, to understand the applicability of the method, it is important to understand its advantages and disadvantages.

Some of the advantages of the SSR method include: 1) you do not have to define a failure surface or search for a minimum failure surface, how the slope fails is a result of the SSR method 2) equations of equilibrium are all satisfied, 3) strains and displacements in the soil and/or rock can be calculated, 4) strains, displacements, axial force and moment distributions in support can be calculated 5) progressive failure can be modeled.

The disadvantages include: 1) Not as widely known or trusted as the limit-equilibrium methods, 2) requires more data such as material modulus, stiffness, plasticity parameters, in-situ stress, boundary conditions etc. 3) Mesh generation and model setup can be difficult and may require a high level of modeling expertise, 4) Limit equilibrium has more material models and is numerically simpler, 5) Finite-element is prone to convergence, tolerance, and numerical instability issues, 6) It is much slower and compute time intensive.

Phase2 version 8 tries to remove a lot of the complexity of defining a finite-element model by directly importing a *Slide* data file, automatically meshing the model, automatically defining in-situ stress states, boundary conditions and material models. Thus limiting the disadvantages talked about above. In the majority of cases, little or no effort is required by the user in order to run a SSR analysis. However, the user must still be aware of what assumptions are made when setting up the finite-element model for a SSR analysis and how the finite-element model is actually created. Below is a description of how a *Slide* file is imported, along with a description of the assumptions made and under what circumstances the user might have to modify the model to accurately calculate the factor of safety. It is important to note that the import of *Slide* files and the automatic model setup is NOT fool-proof. In the majority of cases, the user should only have to import the file and click compute, but be aware that this might not always work.

How Slide Options are Imported into Phase2

The following is a listing of *Slide* version 6 features which are imported into *Phase2*, and those features which are not currently imported into *Phase2*.

Files written with a version of *Slide* prior to 5.0 are not supported but may read correctly depending on what you are trying to model.

Slide Project Settings

Phase2 supports metric and Imperial English units and will properly read *Slide* files with either metric or imperial units (pounds and feet). Other project settings such as failure direction, method, tolerances etc. have no meaning in *Phase2* and are not read. The groundwater setting is read. Sensitivity and probability settings are not read.

Groundwater

Phase2 supports the definition of pore pressures using piezometric lines, R_u , water pressure grids, and steady-state finite element groundwater seepage analysis. The properties and settings for all these techniques are properly read from the *Slide* file during import.

Sensitivity and Probabilistic Analyses

Sensitivity and probabilistic analysis settings from *Slide* are currently NOT imported into *Phase2*. *Phase2* does offer the point estimate method for probabilistic analysis, and applicable parameters (e.g. material property standard deviations) can be copied manually.

Boundaries

The *Slide* external boundary and material boundaries are all read into *Phase2*. The water table is read into *Phase2* but since *Phase2* does not support a specific water table entity, it is converted to a piezometric line with id equal to 1. Piezometric lines are read directly into *Phase2*. Water pressure grids are read into *Phase2*. Tension crack polylines are NOT read into *Phase2*.

Tension Cracks

The explicit modeling of a tension crack region is not directly supported in *Phase2* since no facilities exist in the finite-element method for a zero strength material with possible hydrostatic forces applied to the surface of a tension crack. Consequently, how one models a tension crack zone using a finite-element analysis is open to debate.

One method that has been used successfully (see Verification#27 in the *Phase2* Slope Stability Verification manual), is to represent the soil in the tension crack region as a distributed load applied to the soil underlying the tension crack zone. This works well for dry tension cracks but water filled tension cracks is another issue.

Distributed and Line Loads

Distributed loads (uniform and triangular) and line loads are imported into *Phase2*.

Pseudo-static Seismic Loads

Phase2 supports the import of pseudo-static seismic load coefficients from *Slide*.

Material Properties

The following *Slide* material models are supported: 1) Mohr-Coulomb, 2) Undrained (Constant), 3) Undrained F(datum), 4) Infinite Strength, 5) Shear-Normal Function, 6) Hoek-Brown, 7) Generalized Hoek-Brown, 8) Power Curve.

The following *Slide* material models are not supported: 1) Undrained F(depth), 2) No Strength, 3) Anisotropic Strength, 4) Anisotropic Function, 5) Vertical Stress Ratio, 6) Barton-Bandis, 7) Hyperbolic, 8) Discrete Function, 9) Drained-Undrained.

The Shear-Normal function is supported by fitting a Generalized Hoek-Brown envelope to the discrete data points.

The Power Curve function is supported by converting it to the Generalized Hoek-Brown failure criterion.

The Anisotropic Strength and Anisotropic Function set the material type to Mohr-Coulomb and set the strength as being the minimum of the different directions. Anisotropy in strength is not supported in *Phase2*.

Support and Support Properties

Phase2 will read *Slide* support elements. All support elements in a *Slide* file are read in as *Phase2* bolt elements EXCEPT for geotextiles. Geotextiles are read in as structural interface elements. Structural interfaces have two components: 1) A structural beam element to model the tensile behavior of the geotextile, 2) Two interface elements on either side of the geotextile to model slip between the geotextile and the soil.

Active and passive force application methods for *Slide* support models have no meaning in a *Phase2* finite-element analysis, and are therefore ignored. An equivalent behavior can be defined by setting a Pre-Tensioning force in the *Phase2* bolts.

Slide support models that are imported into *Phase2* are: 1) End Anchored, 2) Geotextiles, 3) Grouted Tieback, 4) Soil Nail.

Support models which are NOT imported: 1) Grouted Tieback (with friction), 2) Micro-Pile.

End anchored or deadman anchors are read in as *Phase2* end-anchored bolts. Peak capacity of the *Phase2* bolt is set to the *Slide* anchor capacity, the residual capacity is set to zero. The bolt spacing is read from the *Slide* file.

Geotextiles will convert to structural interfaces with *Phase2* liner elements being defined as geotextiles with a default tensile modulus and a peak tensile capacity. The peak tensile capacity is read from the *Slide* geotextile support properties. The residual tensile strength is set to zero. The tensile modulus is given a default value equal to 100 times the tensile strength. The user should define the appropriate tensile modulus for the geogrid/geotextile they are using. See the online help for a description of this parameter. If the *Slide* Shear Strength Model for the geotextile-soil interface is linear, the *Phase2* joint interface properties for the structural interface are given a Mohr-Coulomb slip criterion with cohesion and friction angle equal to the adhesion and friction angle defined for the *Slide* geotextile. If the *Slide* Shear Strength Model for the geotextile-soil interface is hyperbolic, the *Phase2* joint interface properties for the structural interface are given a Geosynthetic Hyperbolic slip criterion with adhesion and friction angle equal to the adhesion and friction angle defined for the *Slide* geotextile. Interface normal and shear stiffnesses between the geotextile and the soil are also required. Default values of $K_n=100000\text{KPa/m}$ and $K_s=10000\text{KPa/m}$ are used. These are based on a number of published values and can be changed in the Joint Properties dialog. Material dependant geotextile properties are not read from the *Slide* file but can be manually defined in *Phase2*. *Slide* anchorage methods are supported through the different finite-element mesh end conditions of the structural interface. See the online help for more information on these parameters. Strip coverage is not supported for values other than 100%. You will have to factor the interface and tensile strength properties to account for strip coverage.

Slide Grouted Tiebacks and Soil Nails are both converted to *Phase2* tieback bolts. The only difference between the two is the grouted length. Soil Nails have 100% grouted length. The *Phase2* tieback peak tensile capacity is taken as the minimum of the *Slide* plate capacity and tensile capacity. The residual capacity is set to zero. The bolt spacing is read from the support spacing in the *Slide* file. In the case of tiebacks, the grouted length is properly read. For both *Slide* soil nails and grouted tiebacks, the bond strength is properly read.

Slide Grouted Tiebacks with friction are not properly read into *Phase2*. They are read as *Phase2* tieback bolts but no bond capacity is defined. The user must either define an equivalent bond capacity to the frictional characteristics, thus accounting for the depth of the anchor, or use structural interface elements instead. In the case of structural interface elements, the debonded length of the bolt should be given different material properties than the bonded length. In particular, the debonded length should be given joint stiffness properties (normal and shear) equal to zero. You will require a vertex on the structural interface to separate the bonded from the debonded length.

Micro-piles are not supported in *Phase2*. Piles should be modeled using structural interfaces or liner elements.

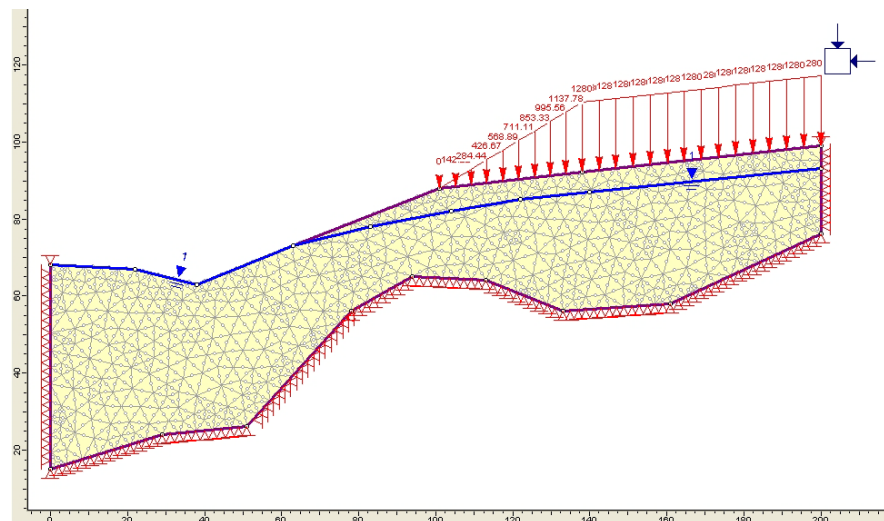
User-defined support properties in *Slide* are not supported in *Phase2*.

Mesh Generation

The complete finite-element mesh is automatically created during the import of the *Slide* file. No user intervention is required. The mesh, by default, will contain approximately 3000 uniformly distributed 6 noded triangular elements.

Boundary Conditions

The import facility automatically determines the top, bottom and sides of the external boundary used in the *Slide* model. The boundary conditions applied to these surfaces are: 1) the top boundary (ground surface) is free to move in the x and y directions, 2) the sides are fixed in the x and y directions (pinned), 3) the bottom surface is fixed in the x and y directions (pinned). The following image shows a typical mesh and boundary conditions after import of a *Slide* model.



Initial Stress and Body Force

By default, each finite-element is given both an initial stress and a body force (self weight). The initial vertical stress is estimated from the weight of the material above the element. *Phase2* automatically determines the ground surface above the element and automatically determines the stress due to the material above the element. The horizontal initial stress is set equal to the vertical stress (hydrostatic stress state). The body force is equal to the unit weight defined for the material in *Slide*. Since *Phase2* allows for only one unit weight, when reading a *Slide* file, the greater of the moist or saturated unit weight is taken.

This system of element loading (the combination of initial stress and body force) is defined in the material properties dialog by defining the Initial Element Loading as being Field Stress & Body Force. Initial Element Loading is one of the more complicated concepts in *Phase2* and it is highly recommended for people who do not understand it, to review the online help on the subject.

Since the initial stress and body force does not define an equilibrium state for a slope (or any non-horizontal ground surface), the material within the slope will deform under the influence of its own self weight and initial stress. In general, the material will deform horizontally away from the slope surface since the initial horizontal stresses are not in equilibrium. The final vertical stress distribution within the slope will be a gravitational stress distribution while the horizontal stress will be due to some unloading and redistribution of stress due to the Poisson effect. When you import a *Slide* file, all imported materials are given a Poisson's ratio of 0.4. If you know your material's Poisson's ratio, you may change the default value inside the *Phase2* material properties dialog.

Horizontal stress plays a very important role in the stability analysis. In general, little is known about the horizontal stress distribution within a soil or rock mass. So assuming that the material has an initial hydrostatic stress state is not unreasonable. This is the assumption made in a large number of the slope stability verification examples. Results from these examples show good agreement with the *Slide* results. If knowledge of the initial vertical and horizontal stress state is known, it should be used in defining the initial stress state for the model.

Ponded Water

In *Phase2*, ponded water is replaced by an equivalent distributed load (pressure) normal to the submerged portion of the external boundary. The distributed load, which varies according to the submerged topology, is defined using a series of "Ponded Water" loads which are oriented normal to the external boundary. When importing a *Slide* file with ponded water, *Phase2* will automatically replace the ponded water by these ponded water distributed loads.

Groundwater Finite-Element Analysis

Both *Slide* and *Phase2* have integrated steady-state unsaturated groundwater modeling capabilities. Thus *Phase2* will read the hydraulic properties (i.e. permeability, unsaturated hydraulic parameters), boundary conditions, and finite-element mesh from the *Slide* data file. By default, if a *Slide* model contains a groundwater mesh, *Phase2* will use this mesh for both stress and groundwater analysis and will not generate a new mesh on import of the *Slide* file. The only exception to this rule is if a distributed load exists in the *Slide* file as well. In this case, the mesh must be created during import but the boundary conditions of the groundwater mesh are preserved.

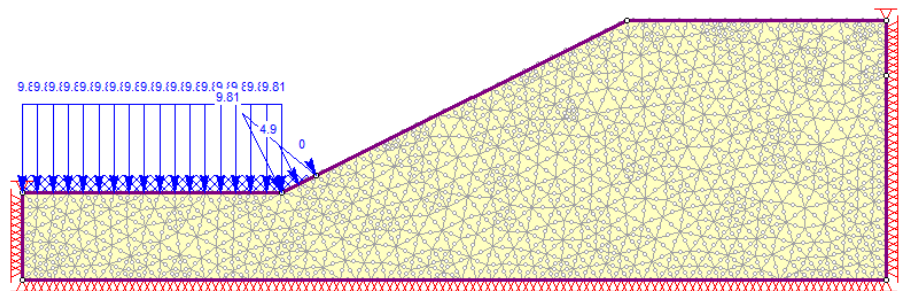
In addition to steady-state analysis *Slide* also offers transient finite element groundwater analysis. Transient groundwater analysis is not currently supported in *Phase2* so transient boundary conditions and material properties are not imported from *Slide*.

Slide File Import / Shear Strength Reduction Example

We will now give a quick demonstration of the Import Slide File option, and the Shear Strength Reduction method in *Phase2*.

Import Slide File

1. In the *Phase2* Model program, select File...Import...Import Slide.
2. Navigate to the Examples > Tutorials folder of your *Phase2* 8.0 installation folder.
3. You will find a Slide file named **Tutorial 09 Slide File.slim**. Open this file.
4. You will see the Slide Import Options dialog. Just select OK in this dialog (leave the default checkbox selections).
5. The file will be imported into *Phase2* and you should see the following model.



Slide file imported into Phase2

NOTE:

- This *Slide* file already included finite element groundwater seepage analysis, therefore the existing groundwater mesh from *Slide* was imported directly into *Phase2*.
- The groundwater boundary conditions in *Slide* defined ponded water at the toe of the slope. As you can see in the above figure, this has been converted into an equivalent distributed load (blue arrows) in *Phase2*.
- As an optional exercise, you can compare the material properties of this model in both *Slide* and *Phase2*. Open this file in *Slide* (assuming you have the *Slide* program). Compare the Material (strength and hydraulic) properties in *Slide* and *Phase2*. You will find that the properties are the same.
- Note that the filename (in *Phase2*) now has a .FEZ filename extension. This is the filename extension used for *Phase2* files.

Compute (with Shear Strength Reduction)

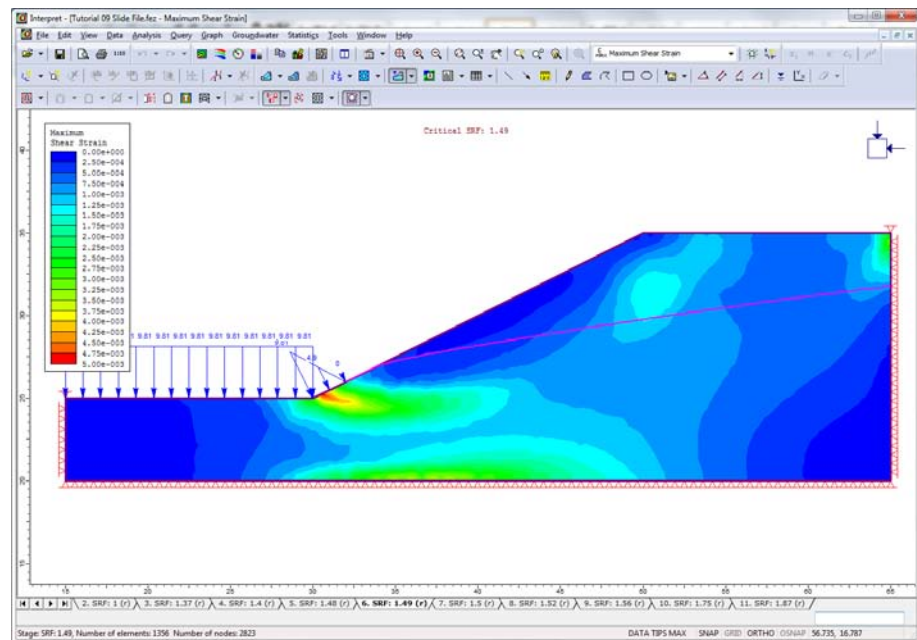
This model does not need any modifications, and can be computed immediately. If this model is computed in *Phase2*, the following computations will be carried out:

1. The finite element groundwater seepage analysis will be run first, to determine pore pressures (this will be virtually instantaneous, you will not notice it).
2. Then the *Phase2* stress analysis will be run, which will include the pore pressures from the groundwater seepage analysis (also very quick for this file).
3. Lastly, the Shear Strength Reduction slope stability analysis will be computed. This may take some time (perhaps a minute or two, depending on the speed of your computer).

Shear Strength Reduction Analysis Results

Select the Interpret button in the *Phase2* Model program, to view the analysis results.

You should see the following:



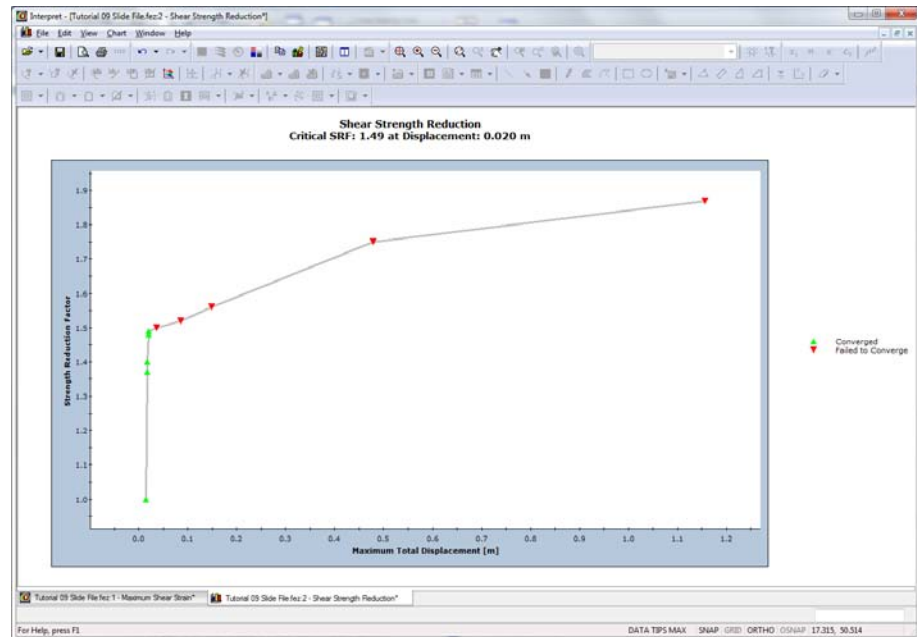
Results of SSR analysis for imported Slide file.

Notice the following:

1. By default, after an SSR slope stability analysis has been performed in *Phase2*, the Maximum Shear Strain contours will be displayed. The Maximum Shear Strain contours highlight the “failure” of the slope at the critical Strength Reduction Factor.
2. The critical SRF represents the Strength Reduction Factor at which the slope becomes unstable (i.e. the stress analysis approaches non-convergence).
3. You will notice that the Stage tabs at the bottom of the screen indicate “SRF: (value)”. Each tab corresponds to ONE iteration of the SSR analysis, using the indicated value of Strength Reduction Factor.
4. By default, the tab with the critical Strength Reduction Factor will be displayed initially. In this case, the critical SRF = 1.49. (Note: this compares with a minimum safety factor slip circle in Slide = 1.52, which is in good agreement). Select the tabs with higher SRF values to view the formation of the slip zone as the shear strength is reduced.
5. By default after an SSR analysis in *Phase2*, only the SSR results are displayed. If you wish to view the regular *Phase2* analysis results (i.e. the results of the stress analysis without applying the Strength Reduction Factor), you must select Data...Stage Settings (in *Phase2* Interpret), and set the Reference Stage = 0 (Not Used). You will then see the results for all stages before the SSR analysis (in this case only the Stage 1 tab) followed by the SSR tabs.

Graphing the Strength Reduction Factor

If you select Graph ... Graph Shear Strength Reduction, you will see the following graph.



This graph summarizes the essential results of the SSR analysis. The Strength Reduction Factor is plotted against the Maximum Displacement (at any point in the model). The critical Strength Reduction Factor corresponds to the point at which the Maximum Displacement shows a sudden increase (i.e. the model becomes unstable).

Importing Surfaces between Slide and Phase2

Before we conclude this quick introduction to SSR analysis with *Phase2*, we will mention a useful feature common to both *Slide* and *Phase2*.

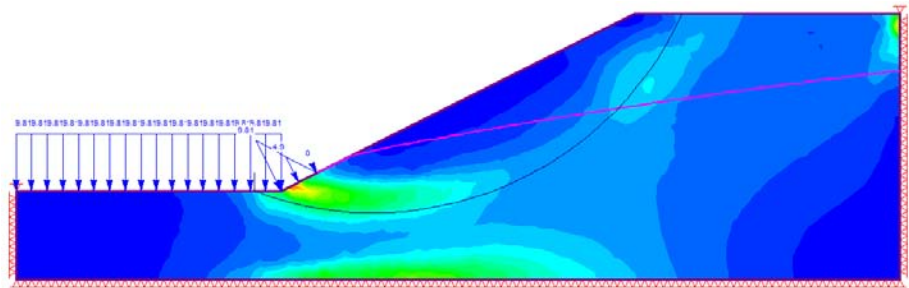
If you wish to compare a limit equilibrium slip surface (determined by *Slide*) with the zone of Maximum Shear Strain contours (after the *Phase2* SSR analysis), you can easily import surfaces (polylines) between *Slide* and *Phase2*.

To import a surface from *Slide* to *Phase2*:

1. Run the model in *Slide*.
2. In the *Slide* Interpret program, right-click the mouse on the critical slip circle/surface.
3. Select Copy (slide modeler format) from the right-click menu.

4. Now, in the *Phase2* Interpret program, go to the Edit menu and select Paste from Slide Interpret.
5. You will see the slip circle/surface from *Slide* Interpret, imported into *Phase2* Interpret. NOTE: the surface is imported as a Polyline Drawing Tool entity.

If you carry out these steps for the current example model, you will see the following:



Notice the critical slip circle (from *Slide*) corresponds approximately to the zone of Maximum Shear Strain contours in *Phase2*.

A similar procedure can be used to import a drawing polyline from the *Phase2* Interpret program, into the *Slide* Model program. In the *Slide* Model program, it can be imported as an actual slip surface, which allows you to run a *Slide* analysis on a surface imported from *Phase2*.

That concludes this tutorial, for more examples of the Shear Strength Reduction method, see the *Phase2* Slope Stability Verification manual, and the accompanying example files, which are installed with the *Phase2* program.